BTS–VI(R)–04.22–0310 Reg. No. | | | | | | | | |

B

B.Tech. Degree VI Semester Regular Examination April 2022

ME 19-205-0605 CAD/CAM

(2019 Scheme)

Time: 3 Hours Maximum Marks: 60

Course Outcome

On successful completion of the course, the students will be able to:

CO1: Understand the basics of Computer Aided Design (CAD), geometric modelling, automation, flow line analysis, and line balancing

CO2: Understand the structure and special design features of CNC machines. Learn manual CNC programming and computer-aided programming technique like APT.

CO3: Understand the basic configurations, components, applications and the programming of robots.

CO4: Learn the fundamentals of the advanced concepts in automation and manufacturing like DNC, CIM, FMS, and CAE and the basic concepts of AI and expert systems for manufacturing automation. Learn the basic concepts and applications of Rapid Prototyping.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 –Analyze, L5 – Evaluate, L6 – Create

PO – Programme Outcome

PART A

(Answer *ALL* questions)

		(8 × 3 = 24)	Marks	BL	CO	PO
I.	(a)	List down the steps in the general design process (Shigley's design process) and explain the use of computers in each step. Indicate the name of a CAD package used in each step.	3	L2	1	1,2,5
	(b)	Explain flexible automation. Discuss the type of industry where flexible automation is most desirable.	3	L3	1	1,2,5
	(c)	What are the components of a CNC machine? Explain with proper illustration.	3	L1	2	1,5
	(d)	What is the stick-slip phenomenon in machine tool guideways? How can the stick-slip be eliminated in guideways?	3	L2	2	1,2,3
	(e)	What are the features of the robots used for the assembly operation? Suggest the best robotic configuration for assembly with justification.	3	L4	3	1,2,4
	(f)	Explain the textual language programming of robots. List down and explain any four AML commands.	3	L1	3	1,2,5
	(g)	Write short notes on CIM, FMS and DNC.	3	L1	4	1,4,5
	(h)	Explain the challenges in the additive manufacturing of metals.	3	L5	4	1,2,5,6

PART B

(4 × 12 = 48)

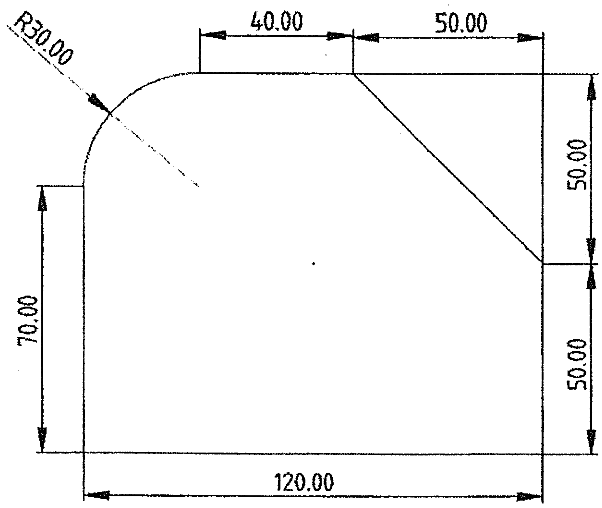
			Marks	BL	CO	PO
II.	(a)	Explain the following techniques in solid modeling: (i) Extrusion (ii) Revolution (iii) Lofting (iv) Sweeping.	6	L1	1	1,2,5
	(b)	Rotate the square with coordinates (1,1), (3,1), (3,3) and (1,3) about the point (2, 2). Use homogeneous coordinates and concatenation of translation and rotation matrices.	6	L4	1	1,2,5

OR

III.	(a)	What are the different levels of automation? Explain each with a proper illustration.	4	L1	1	1,4,5
	(b)	A new small electrical appliance is to be assembled on a production flow line. The total job of assembling the product has been divided into minimum rational work elements. The table shows the elemental times and their immediate predecessors. The cycle time is 1 minute and the line efficiency is 100%. (i) Construct the precedence diagram from the data provided on work elements (ii) Use the largest-candidate rule to assign the work elements to stations. Find out the balance delay.	8	L4	1	1,2,4,5

Activity No	1	2	3	4	5	6	7	8	9	10	11	12
Element Time T_{ej}	0.2	0.4	0.7	0.1	0.3	0.11	0.32	0.6	0.27	0.38	0.5	0.12
Must be preceded by	-	-	1	1,2	2	3	3	3,4	6,7,8	5,8	9,10	11

IV.	(a)	How are NC/CNC machines classified? Explain each type with appropriate illustrations.	4	L1	2	1,2,5
	(b)	Write a CNC program to mill the profile shown in the figure from a plate of appropriate size. The thickness of the plate is *5 mm*. Use an HSS endmill, $\phi 20$ mm for milling the profile.	8	L3	2	1,2,5



All dimensions in mm

OR

(Continued)

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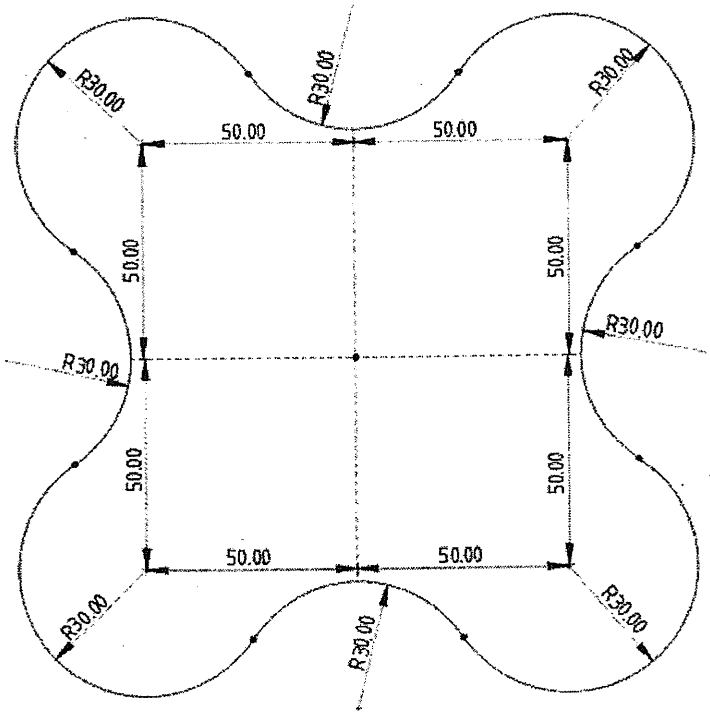
V.	(a)	Compare ball screws with lead screws in terms of efficiency, performance and other features. How is pre-loading of ball screws done?	4	L5	2	1,2,3,5
	(b)	Write an APT programme for generating CNC code for the profile given below from a mild steel plate of 170 × 170 × 5 mm size. Use a carbide endmill of 20 mm diameter. All dimensions are in mm.	8	L3	2	1,2,5



VI.	(a)	Draw the work envelope of (i) a SCARA (ii) an articulated robot.	6	L2	3	1,2,3
	(b)	Explain in brief the control loops used in robotic systems.	6	L1	3	1,2,3,5
		OR				
VII.		Make a conceptual design of a gantry robot (rectangular/cartesian configuration) for loading and unloading a CNC machining centre. Explain the configuration of the system and the components with the help of neat diagrams.	12	L6	3	1,2,3,5
VIII.	(a)	What is adaptive (intelligent) manufacturing? Evaluate different strategies used in adaptive manufacturing.	6	L5	4	1,2,5
	(b)	Discuss the use of expert systems in manufacturing.	6	L3	4	1,4
		OR				
IX.	(a)	Explain selective laser sintering (SLS) with the help of neat figures.	6	L2	4	1,5
	(b)	Explain selective laser melting (SLM)with the help of neat figures.	6	L2	4	1,5

$L1-30\%$ $L2-22\%$ $L3-18\%$ $L4=13\%$ $L5-13\%$ $L6-4\%$

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